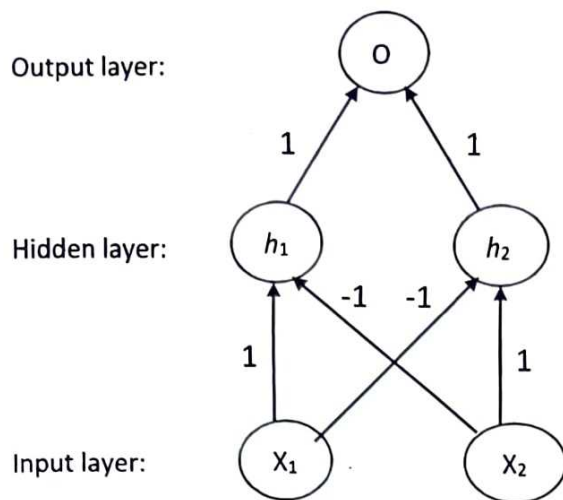




Sigmoid activation function

Learning rate 0.7

Trace the algorithm.

https://www.tinkershop.net/ml/sigmoid_calculator.html

repeat

for each e in examples do

for each node j in the input layer do $a_j \leftarrow x_j[e]$

for $\ell = 2$ to M do

$in_i \leftarrow \sum_j W_{j,i} a_j$

$a_i \leftarrow g(in_i)$

for each node i in the output layer do

$\Delta_i \leftarrow g'(in_i) \times (y_i[e] - a_i)$

for $\ell = M - 1$ to 1 do

for each node j in layer ℓ do

$\Delta_j \leftarrow g'(in_j) \sum_i W_{j,i} \Delta_i$

for each node i in layer $\ell + 1$ do

$W_{j,i} \leftarrow W_{j,i} + \alpha \times a_j \times \Delta_i$

until some stopping criterion is satisfied

$$\Delta O = O'(y - O)$$

$$\Delta h1 = h'(W)(\Delta O)$$

$$W' = \alpha \times \text{input} \times \Delta$$

XOR FFNet

X1	X2	y	h_1	h_2	h'_1	h'_2	O	O'	ΔO	$\Delta h1$	$\Delta h2$	W'_{h1-O}	W'_{h2-O}
0	0	0	0.500	0.500	0.250	0.250	0.731	0.197	-0.144	-0.036	-0.036	-0.0504	-0.0504
0	1	1	0.269	0.731	0.197	0.197	0.731	0.197	0.053	0.010	0.010	0.0100	0.0271
1	0	1	0.731	0.269	0.197	0.197	0.731	0.197	0.053	0.010	0.010	0.0271	0.0100
1	1	0	0.500	0.500	0.250	0.250	0.731	0.197	-0.144	-0.036	-0.036	-0.0504	-0.0504

X1	X2	W_{h1-O}	W_{h2-O}	W'_{x1-h1}	W'_{x1-h2}	W_{x1-h1}	W_{x1-h2}	W'_{x2-h1}	W'_{x2-h2}	W_{x2-h1}	W_{x2-h2}
0	0	0.9496	0.9496	0.000	0.000	1.000	-1.000	0.000	0.000	-1.000	1.000
0	1	1.0100	1.0271	0.000	0.000	1.000	-1.000	0.007	0.007	-0.993	1.007
1	0	1.0271	1.0100	0.007	0.007	1.007	-0.993	0.000	0.000	-1.000	1.000
1	1	0.9496	0.9496	-0.0252	-0.0252	0.9748	-1.0252	-0.0252	-0.0252	-1.0252	0.9748